

Kai Neuhaus

PhD, BSc, Dipl.-Ing.

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Postdoctoral Fellow

Optical Coherence Tomography, Biomedical Imaging

EXPERIENCE

OREGON HEALTH & SCIENCE UNIVERSITY | POSTDOCTORAL FELLOW

Since Jan 2022 | Portland, OR

- Casey Eye Institute, Center for Ophthalmic Optics & Lasers
- Design and investigation of advanced optical medical imaging methods to enhance Optical Coherence Tomography technology to prevent blindness
- Collaborative research with Siloam Vision Inc.

NATIONAL UNIVERSITY OF IRELAND, GALWAY | POSTDOCTORAL RESEARCHER

Sep 2020 – Feb 2022 | Galway, Ireland

- Tissue Optics & Microcirculation Imaging
- Time and Spectral Domain Optical Coherence Tomography
- Investigation of skin imaging methods for cancer detection

RESEARCH ASSISTANT

Jan 2020 – Sep 2020 | Galway, Ireland

- Software development for biomedical image analysis

COMPACT IMAGING IRELAND LTD | APPLICATION ENGINEER

Sep 2018 – Jan 2020 | Limerick, Ireland

- Co-founding the Ireland branch of California-based Compact Imaging Inc.
- Leading the engineering department
- Research and development of miniature optical coherence tomography systems
- Management of ISO and medical documentations

NCLA - NATIONAL CENTRE OF LASER APPLICATIONS | RESEARCH ENGINEER

Sep 2012 – Mar 2013 | Galway, Ireland

RESEARCH ENGINEER

- Development of a high-resolution optical temperature sensor

Feb 2011 – Dec 2011 | Galway, Ireland

RESEARCH ASSISTANT

- Laser processing and laser calibration

T-SYSTEMS | PROJECT DEVELOPER

1996 – 2003 | Chemnitz, Germany

- Software quality management and development
- Large-scale Java programming environment
- Consultancy to IBM Germany, RWE AG (energy provider), and mobile phone accounting systems

RESEARCH PROJECTS

OREGON HEALTH & SCIENCE UNIVERSITY, CASEY EYE INSTITUTE REAL-TIME LINE-FIELD OPTICAL COHERENCE TOMOGRAPHY FOR CELLULAR RESOLUTION IMAGING OF BIOLOGICAL TISSUE

Postdoctoral research project to advance ophthalmic imaging methods to prevent blindness for newborn infants. Investigation of machine learning methods for data analysis.

NATIONAL UNIVERSITY OF IRELAND, GALWAY TISSUE OPTICS & MICROCIRCULATION IMAGING (TOMI) LABORATORY, GALWAY, IRELAND

BIOELECTRICAL IMPEDANCE SPECTROSCOPY (BIS)

Investigation on human skin sodium detection using impedance spectroscopy as a biomarker to detect hypertension risks

NATIONAL UNIVERSITY OF IRELAND, GALWAY TISSUE OPTICS & MICROCIRCULATION IMAGING (TOMI) LABORATORY, GALWAY, IRELAND

MULTIPLE REFERENCE OPTICAL COHERENCE TOMOGRAPHY

Graduate research in hardware design and development, image and signal analysis on low-cost miniature ophthalmic screening system. Conference presentations at Photonics West, San Francisco, CA

EDUCATION

NATIONAL UNIVERSITY OF IRELAND, GALWAY 2013 - 2018 | PHD IN PHYSICS

<https://www.universityofgalway.ie/>

Enhanced studies in optical design, optical coherence tomography, and biomedical imaging

GALWAY-MAYO INSTITUTE OF TECHNOLOGY 2009 - 2012 | BSc (FIRST CLASS HONORS)

PHYSICS AND INSTRUMENTATION

<https://www.atu.ie/>

Measurement and detection systems, Quantum Physics, Metrology, Spectroscopy, XRF, AFM, SEM, system modeling

TECHNISCHE UNIVERSITÄT CHEMNITZ 1991 - 1995 | DIPL.-ING. PHYSIK UND MIKROSYSTEME

<https://www.tu-chemnitz.de/> Chemnitz, Germany

Design and FEM simulation of silicon-based microsystems

INTERN AT BOSCH GMBH, FEM AND SIGNAL ANALYSIS

AWARDS & RECOGNITIONS

Sep 2016 | SFI Poster Prize - Advanced Laser Technologies Conference, Ireland

Feb 2016 | Travel Scholarship Cycle 2, SPIE (<https://www.spie.org/>)

Nov 2012 | Student of the year supporting peer-assisted learning, GMIT, Ireland

PROFESSIONAL MEMBERSHIPS

SPIE | <https://www.spiedigitallibrary.org/profile/Kai.Neuhaus-2152>

OSA | <https://www.optica.org/>

IOP | <https://www.iop.org/>

IEEE | <https://ieeexplore.ieee.org/author/37086095216>

LANGUAGES

PROGRAMMING

Java • LabView • Shell • Matlab • Python • $\LaTeX$ • Lisp • ELisp

C • C++ • Zemax Macro

Frameworks/Platforms:

Zemax • SQL • Git • SVN • Emacs • Windows • Linux • MacOS • FreeCAD

Familiar:

Rust • Julia •

SPOKEN & WRITTEN

Native fluency:

German, English

PUBLICATIONS & CONTRIBUTIONS

- Dsouza, Roshan, Hrebesh Subhash, Kai Neuhaus, Josh Hogan, Carol Wilson and Martin Leahy (2014b). "Dermascope guided multiple reference optical coherence tomography". In: Biomed. Opt. Express 5, pp. 2870–2882. doi: **10.1364/BOE.5.002870**.
- Dsouza, Roshan, Hrebesh M. Subhash, Kai Neuhaus, Josh Hogan, Carol Wilson and Martin Leahy (2015). "3D nondestructive testing system with an affordable multiple reference optical-delay-based optical coherence tomography". In: Appl. Opt., AO 54.18, pp. 5634–5638. issn: 1539-4522. doi: **10.1364/AO.54.005634**.
- Dsouza, Roshan, Hrebesh Subhash, Kai Neuhaus, Ramakrishna Kantamneni, Paul M. McNamara, Josh Hogan, Carol Wilson and Martin Leahy (2016). "Assessment of curing behavior of light-activated dental composites using intensity correlation based multiple reference optical coherence tomography". In: Lasers Surg. Med. 48.1, pp. 77–82. issn: 1096-9101. doi: **10.1002/lsm.22447**.
- Butler, T. P., S. Slepneva, P. M. McNamara, K. Neuhaus, D. Goulding, M. Leahy and G. Huyet (2017). "Real-Time Experimental Measurement of Swept Source VCSEL Properties Relevant to OCT Imaging". In: IEEE Photonics Journal 9.5, pp. 1–10. doi: **10.1109/JPHOT.2017.2752644**.
- Dsouza, Roshan, Hrebesh Subhash, Kai Neuhaus, Paul M. McNamara, Josh Hogan, Carol Wilson and {and} Martin J. Leahy (2017). "Feasibility study of phase-sensitive imaging based on multiple reference optical coherence tomography". In: Chin. Opt. Lett., COL 15.9, p. 090007.
- Neuhaus, Kai, Seán O’Gorman, Paul M. McNamara, Sergey A. Alexandrov, Josh Hogan, Carol Wilson and Martin J. Leahy (2017). "Simultaneous en-face imaging of multiple layers with multiple reference optical coherence tomography". In: JBO 22.8. tex.ids: neuhausSimultaneousEnfaceImaging2017a, pp. 1–7. issn: 1083-3668, 1560-2281. doi: **10.1117/1.JBO.22.8.086006**.
- Neuhaus, Kai, Paul M. McNamara, Sergey Alexandrov, Sean O’Gorman, Josh Hogan, Carol Wilson and Martin J. Leahy (2018). "Performance Review of Multiple Reference Versus Time Domain Optical Coherence Tomography". In: IEEE Photonics Journal 10. tex.ids: neuhausPerformanceReviewMultiple2018a, p. 2828419. doi: **10.1109/JPHOT.2018.2828419**.
- Neuhaus, Kai (2019b). "The multi-spectral signal properties of multiple reference optical coherence tomography". In: PhD Thesis. <http://hdl.handle.net/10379/15318>.
- Alexandrov, Sergey, Anand Arangath, Yi Zhou, Mary Murphy, Niamh Duffy, Kai Neuhaus, Georgina Shaw, Ryan McAuley and Martin Leahy (2021). "Accessing depth-resolved high spatial frequency content from the optical coherence tomography signal". In: Sci Rep 11.1. Bandiera_abtest: a Cc_license_type: cc_by Cg_type: Nature Research Journals Number: 1 Primary_atype: Research Subject_term: Imaging and sensing;Imaging techniques;Optical imaging;Optical techniques Subject_term_id: imaging-and-sensing;imaging-techniques;optical-imaging;optical-techniques, p. 17123. issn: 2045-2322. doi: **10.1038/s41598-021-96619-7**.
- James, Soorya, Kai Neuhaus, Mary Murphy and Martin Leahy (2021). "Contrast agents for photoacoustic imaging: a review of stem cell tracking". In: Stem Cell Res Ther 12.1, p. 511. issn: 1757-6512. doi: **10.1186/s13287-021-02576-3**.
- Arangath, Anand, Kai Neuhaus, Sergey Alexandrov and Martin Leahy (2022). "Evaluation of Signal Degradation Due to Birefringence in a Multiple Reference Optical Coherence Tomography System With Polarization-Based Balanced Detection". In: IEEE Photonics Journal 14.4. Conference Name: IEEE Photonics Journal, pp. 1–12. issn: 1943-0655. doi: **10.1109/JPHOT.2022.3189809**.
- Dey, Rajib, Yi Zhou, Kai Neuhaus, Sergey Alexandrov, Andrew Nolan, Ting-Chiao Chang and Martin Leahy (2022). "Simple Characterization Scheme for Optical Coherence Tomography Systems With Application to a Commercial and a Near-Isometric Resolution Fibre-Based System". In: IEEE Photonics Journal 14.1. Conference Name: IEEE Photonics Journal, pp. 1–11. issn: 1943-0655. doi: **10.1109/JPHOT.2021.3135058**.
- Khan, Shanjida, Kai Neuhaus, Omkar Thaware, Shuibin Ni, Myeong Jin Ju, Travis Redd, David Huang and Yifan Jian (2022). "Corneal imaging with blue-light optical coherence microscopy". In: Biomed. Opt. Express, BOE 13.9, pp. 5004–5014. issn: 2156-7085. doi: **10.1364/BOE.465707**.
- Arangath, Anand, Niamh Duffy, Sergey Alexandrov, Soorya James, Kai Neuhaus, Mary Murphy and Martin Leahy (2023). "Nanosensitive optical coherence tomography for detecting structural changes in stem cells". In: Biomed. Opt. Express, BOE 14.4, pp. 1411–1427. issn: 2156-7085. doi: **10.1364/BOE.485082**.
- Arangath, Anand, Kai Neuhaus, Sergey Alexandrov and Martin Leahy (2023). "Digitally balanced detection scheme in multiple reference optical coherence tomography". In: Opt. Continuum, OPTCON 2.2, pp. 397–412. issn: 2770-0208. doi: **10.1364/OPTCON.476099**.
- Khan, Shanjida, Kai Neuhaus, Omkar C. Thaware, Mini Aga, Alireza Karimi, Mary J. Kelley, Travis Redd, Ted S. Acott, David Huang and Yifan Jian (2023). "Visible-light optical coherence microscopy". In: Invest. Ophthalmol. Vis. Sci. 64.8, pp. 1966–1966. issn: 1552-5783.
- Neuhaus, Kai, Shanjida Khan, Omkar Thaware, Shuibin Ni, Mini Aga, Yali Jia, Travis Redd, Siyu Chen, David Huang and Yifan Jian (2024). "Real-time line-field optical coherence tomography for cellular resolution imaging of biological tissue". In: Biomed. Opt. Express, BOE 15.2, pp. 1059–1073. issn: 2156-7085. doi: **10.1364/BOE.511187**.
- Neuhaus, Kai, Shanjida Khan, Omkar C Thaware, Shuibin Ni, Mini Aga, Yali Jia, Travis K. Redd, Siyu Chen, David Huang and Yifan Jian (2024). "In Vivo Continuous High-speed Line-Field Optical Coherence Tomography corneal and retinal imaging". In: Investigative Ophthalmology & Visual Science 65.7, OD53. issn: 1552-5783.
- Zhou, Yuxiang, Mingliang Zhou, Bo Wang, Xiaoting Yin, Jing Bai, Shuai Wang, Kai Neuhaus, Bernhard Baumann, Yifan Jian and Pengfei Zhang (2025). "Development of a High-Speed Swept-Source OCT/OCTA/ORG System for Structural and Functional Imaging of the Living Mouse Retina". In: IEEE Transactions on Instrumentation and Measurement 74, pp. 1–12. issn: 1557-9662. doi: **10.1109/TIM.2025.3606015**.

- Dsouza, Roshan, Hrebesh Subhash, Kai Neuhaus, Josh Hogan, Carol Wilson and Martin Leahy (2014a). "Dermascope assisted interactive patient interface for multiple reference optical coherence tomography". In: Proc. SPIE. Vol. 8935. Proc. SPIE, OB-9. doi: **10.1117/12.2038183**.
- Subhash, Hrebesh M., Kai Neuhaus, Roshan Dsouza, Josh Hogan, Carol Wilson and Martin J. Leahy (2014). "Smartphone based Multiple Reference Optical coherence tomography (MRO™) system". In: Biomedical Optics 2014 (2014), paper BT3A.72. Biomedical Optics. Optical Society of America, BT3A.72. doi: **10.1364/BIOMED.2014.BT3A.72**.
- Subhash, Hrebesh M., Sean O’Gorman, Kai Neuhaus and Martin Leahy (2014). "High resolution coherence domain depth-resolved nailfold capillaroscopy based on correlation mapping optical coherence tomography". In: Proc. SPIE. Optical Diagnostics and Sensing XIV. Vol. 8951. SPIE, pp. 895106–895106–5. doi: **10.1117/12.2038550**.
- Dsouza, Roshan, Molly Subhash Hrebesh, Kai Neuhaus, Josh Hogan, Carol Wilson and M. J. Leahy (2015). "A novel low cost 3-D nondestructive testing system based on multiple reference optical coherence tomography". In: SPIE Photonics West. Vol. 9314. SPIE, pp. 9312–134.
- Neuhaus, Kai, Hrebesh Subhash, Roshan Dsouza, Josh Hogan, Carol Wilson and Martin Leahy (2015a). "Characterization of light distribution and optimization of detector position for multiple reference optical coherence tomography". In: Optics and Biophotonics in Low-Resource Settings. Proc. SPIE. Vol. 9314. SPIE, 93140F–93140F–9. doi: **10.1117/12.2079744**.
- (2015b). "Signal simulation and signal processing for multiple reference optical coherence tomography". In: ptical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XIX. Proc. SPIE. Vol. 9312. SPIE, pp. 931236–931236–7. doi: **10.1117/12.2080994**.
- Leahy, M. J., C. Wilson, J. Hogan, Peter O’Brien, R. Dsouza, K. Neuhaus, D. Bogue, H. Subhash, Colm O’Riordan and Paul M. McNamara (2016). "The how and why of a \$10 optical coherence tomography system". In: vol. 9697, 96970T–96970T–3. doi: **10.1117/12.2213465**.
- Neuhaus, Kai, Hrebesh Subhash, Sergey Alexandrov, Roshan Dsouza, Josh Hogan, Carol Wilson, Martin Leahy, Svetlana Slepneva and Guillaume Huyet (2016). "The impact of relative intensity noise on the signal in multiple reference optical coherence tomography". In: Optics and Biophotonics in Low-Resource Settings II. Optics and Biophotonics in Low-Resource Settings II. Vol. 9699. tex.ids: neuhausImpactRelativeIntensity2016a. International Society for Optics and Photonics, 96990P. doi: **10.1117/12.2211404**.
- O’Gorman, Sean, Paul M. McNamara, Roshan Dsouza, Kai Neuhaus, Josh Hogan, Carol Wilson and Martin J. Leahy (2016). "Evaluation of a polarization sensitive multiple reference optical coherence tomography system". In: Optics and Biophotonics in Low-Resource Settings II. Optics and Biophotonics in Low-Resource Settings II. Vol. 9699. International Society for Optics and Photonics, 96990E. doi: **10.1117/12.2211094**.
- Neuhaus, Kai, Sergey Alexandrov, Roshan Dsouza, Sean O’Gorman, Hrebesh M. Subhash, Josh Hogan, Carol Wilson, Paul McNamara and Martin Leahy (2017). "Comparing an FFT filter for multiple reference optical coherence tomography (MR-OCT) with an Chebychev and an elliptic filter". In: Optics and Biophotonics in Low-Resource Settings III. Optics and Biophotonics in Low-Resource Settings III. Vol. 10055. International Society for Optics and Photonics, pp. 100550X–100550X–1. doi: **10.1117/12.2253265**.
- Arangath, Anand, Kai Neuhaus, Yi Zhou, Sergey Alexandrov, Josh Hogan, Carol Wilson and Martin Leahy (2020). "Optimization of SNR, sensitivity values by evaluating the relative spacing between the partial mirror and scanning reference mirror for multiple reference optical coherence tomography". In: Optics and Biophotonics in Low-Resource Settings VI. Optics and Biophotonics in Low-Resource Settings VI. Vol. 11230. International Society for Optics and Photonics, p. 112300X. doi: **10.1117/12.2550331**.
- Dey, Rajib, Yi Zhou, Nandan Das, Sergey Alexandrov, Andrew Nolan, Kai Neuhaus and Martin Leahy (2020). "Development of HR-SD-OCT system using supercontinuum light source and its application in detecting nanoscale changes". In: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXIV. Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXIV. Vol. 11228. SPIE, pp. 121–125. doi: **10.1117/12.2543398**.
- Zhou, Yi, Sergey Alexandrov, Andrew Nolan, Rajib Dey, Nandan Das, Kai Neuhaus and Martin Leahy (2020). "Application of over-sampling nano-sensitive optical coherence tomography for monitoring corneal internal structural changes in corneal cross-linking". In: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXIV. Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXIV. Vol. 11228. SPIE, pp. 74–78. doi: **10.1117/12.2547317**.
- Arangath, Anand, Niamh Duffy, Sergey Alexandrov, Soorya James, Kai Neuhaus, Mary Murphy and Martin Leahy (2022). "Nanosensitive Optical Coherence Tomography for Tracking of Mesenchymal Stem Cells Labelled With Gold Nanostars". In: Conference on Lasers and Electro-Optics (2022), paper AM5I.3. CLEO: Applications and Technology. Optica Publishing Group, AM5I.3. doi: **10.1364/CLEO_AT.2022.AM5I.3**.
- Huang, David, Shanjida Khan, Kai Neuhaus, Omkar Thaware, Alireza Karimi, Mary Kelley, Travis Redd, Ted Acott and Yifan Jian (2023a). "Visible-light optical coherence microscopy". In: Optical Coherence Imaging Techniques and Imaging in Scattering Media V. Optical Coherence Imaging Techniques and Imaging in Scattering Media V. Vol. 12632. SPIE, pp. 78–82. doi: **10.1117/12.2675364**.
- (2023b). "Visible-light optical coherence microscopy". In: Optical Coherence Imaging Techniques and Imaging in Scattering Media V (2023), paper 126320Q. European Conference on Biomedical Optics. Optica Publishing Group, 126320Q. doi: **10.1117/12.2675364**.

- Khan, Shanjida, Kai Neuhaus, Omkar Thaware, Mini Aga, Alireza Karimi et al. (2023). "Visible-light optical coherence microscopy for corneal imaging". In: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXVII. Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXVII. Vol. 12367. SPIE, pp. 87–90. doi: **10.1117/12.2652252**.
- Khan, Shanjida, Kai Neuhaus, Omkar Thaware, Shuibin Ni, Myeong Ju, Travis Redd, David Huang and Yifan Jian (2023). "Blue-light optical coherence microscopy". In: APS March Meeting Abstracts. Vol. 2023. ADS Bibcode: 2023APS..MART44005K, T44.005.
- Neuhaus, Kai, Shuibin Ni, Shanjida Khan, Omkar Tharawe, Siyu Chen, David Huang and Yifan Jian (2023). "Real-time line-field OCT using low-cost high-speed camera". In: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXVII. Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXVII. Vol. 12367. SPIE, pp. 61–67. doi: **10.1117/12.2652656**.
- (2024). "In-vivo real-time imaging of the human cornea and retina with high-speed line-field optical coherence tomography". In: Ophthalmic Technologies XXXIV. Ophthalmic Technologies XXXIV. Vol. 12824. SPIE, pp. 119–124. doi: **10.1117/12.3001773**.